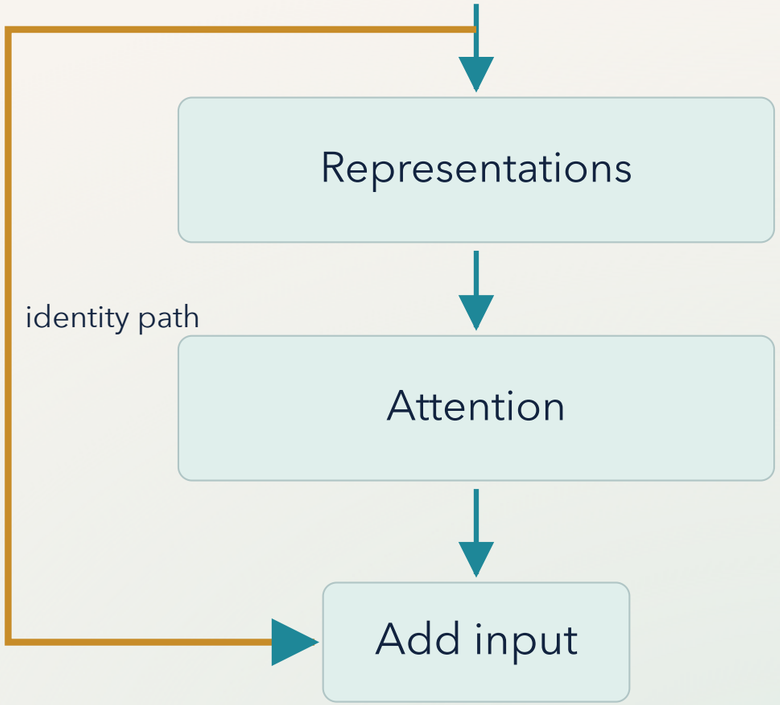
Build a model around attention

A Transformer block combines information mixing with local transformation and stable update paths.



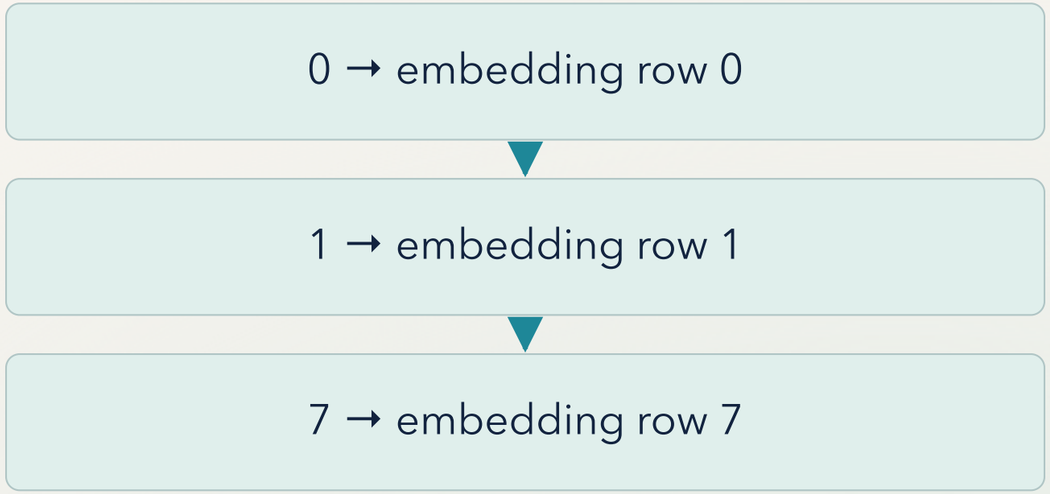
Then: normalized feed-forward update + residual

Mechanism diagram / illustrative teaching values

Tokens are discrete input units

Token IDs index learned vectors; IDs do not encode numeric magnitude.

> `token ID → embedding lookup → vector`

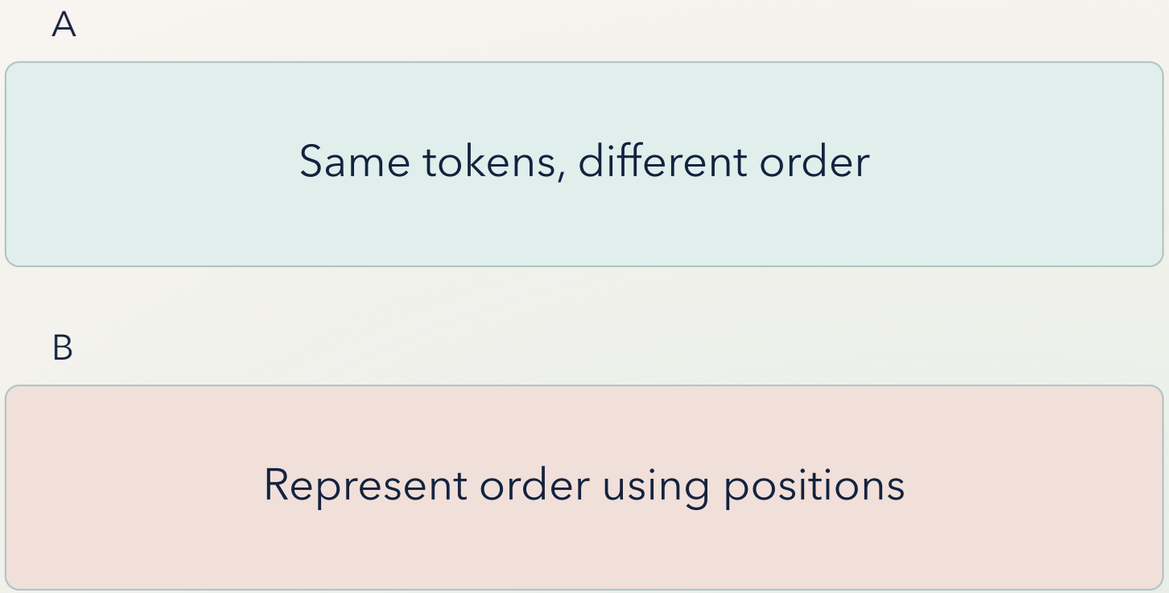


Mechanism diagram / illustrative teaching values

Position information supplies order

Unmasked attention alone is permutation-equivariant when positions are absent.

$$\text{Attention}(PX) = P\,\text{Attention}(X)$$

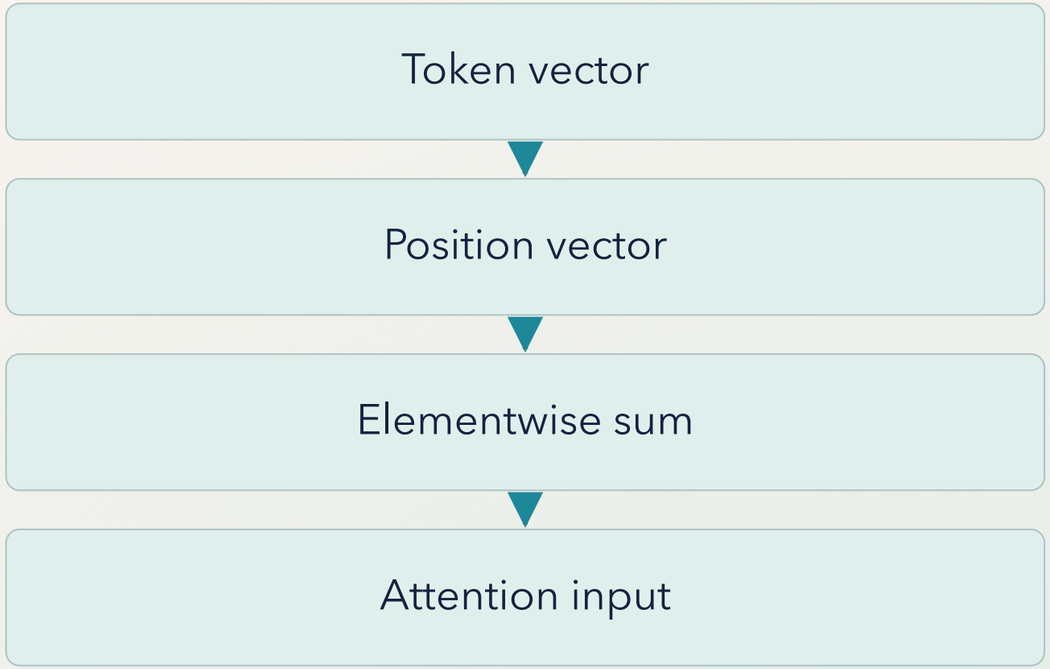


Mechanism diagram / illustrative teaching values

There are several position mechanisms

The teaching model adds learned position vectors; the original Transformer used sinusoidal encodings.

> z_t = token_embedding(id_t) + position_embedding(t)

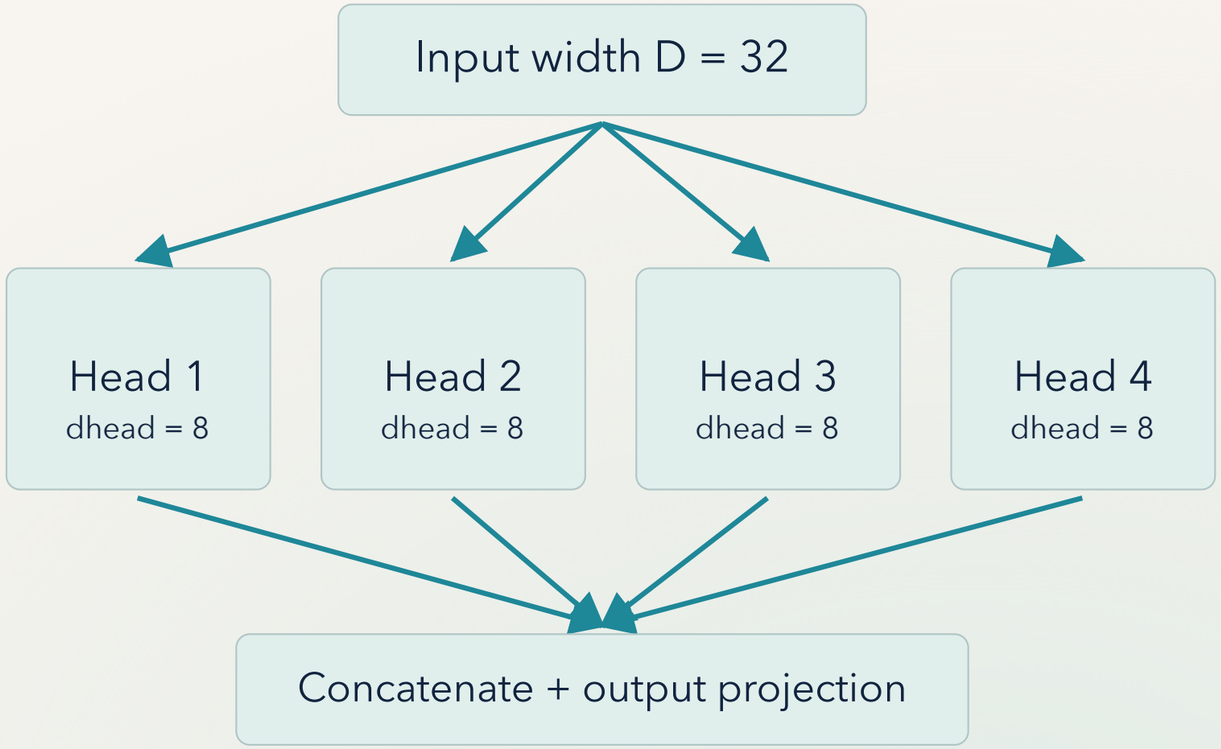


Mechanism diagram / illustrative teaching values

Multiple heads learn multiple projections

Each head routes information in its own learned query/key/value subspace.

> $D = H \times d_{head}$

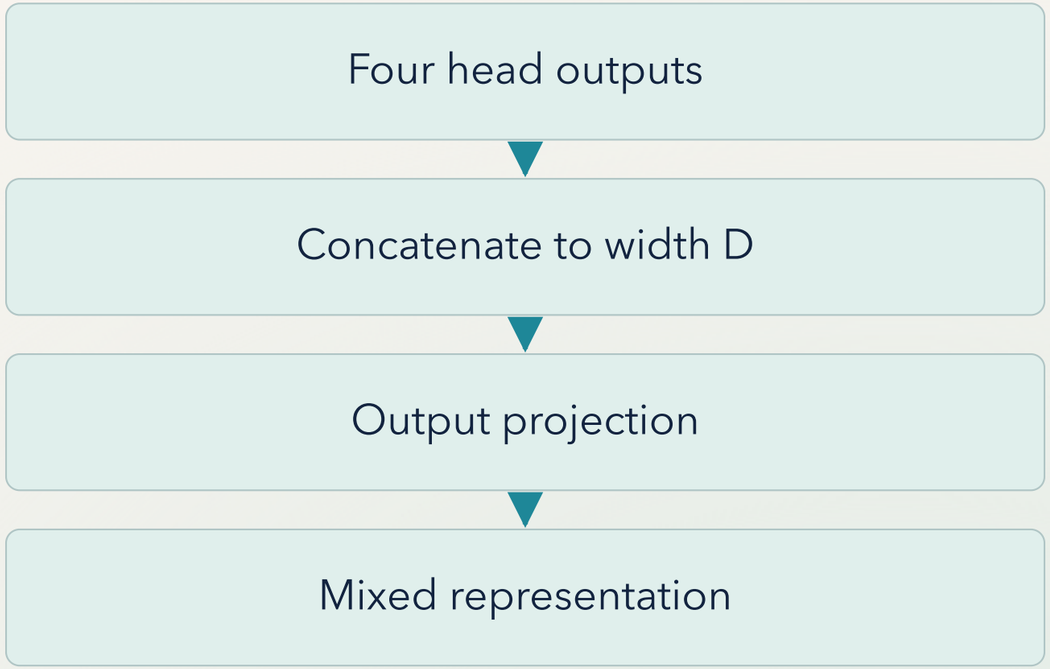


Mechanism diagram / illustrative teaching values

Concatenate heads, then project

Head outputs are joined and mixed by a learned output projection.

$$\text{MultiHead} = \text{Concat}(\text{head}_1, \ldots, \text{headH})\,W^O$$

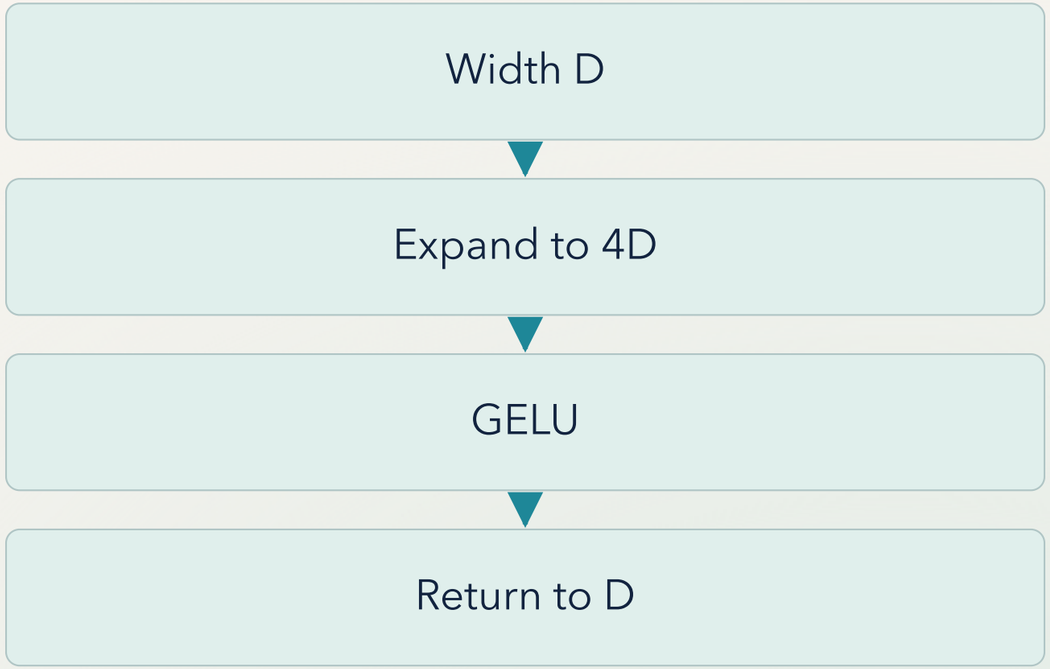


Mechanism diagram / illustrative teaching values

The feed-forward network transforms each token

The same nonlinear MLP is applied independently at each position.

$$\mathrm{FFN}(x) = W_2\,\mathrm{GELU}(W_1 x + b_1) + b_2$$

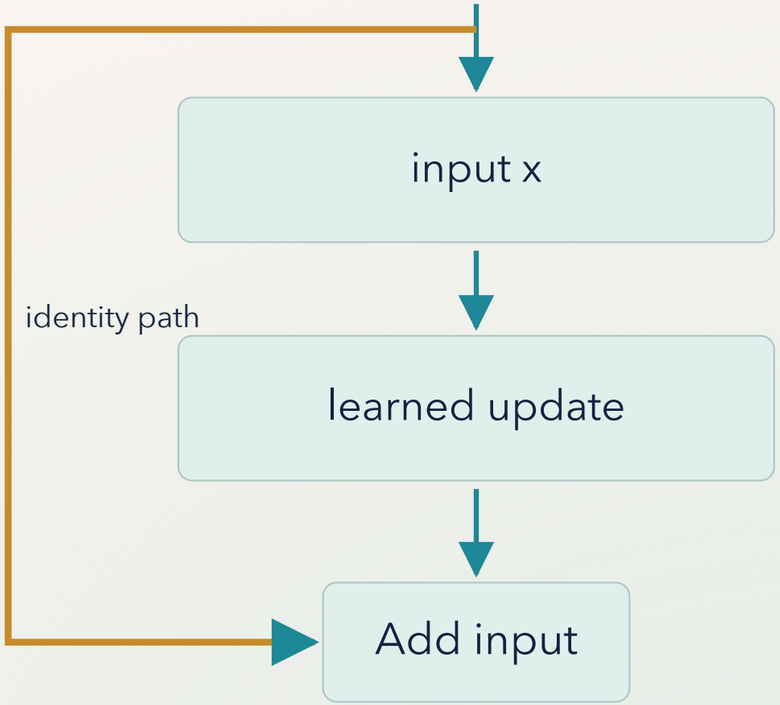


Mechanism diagram / illustrative teaching values

Residual paths preserve an identity route

A sublayer adds a learned update to its input representation.

```
y = x + Sublayer(x)
```

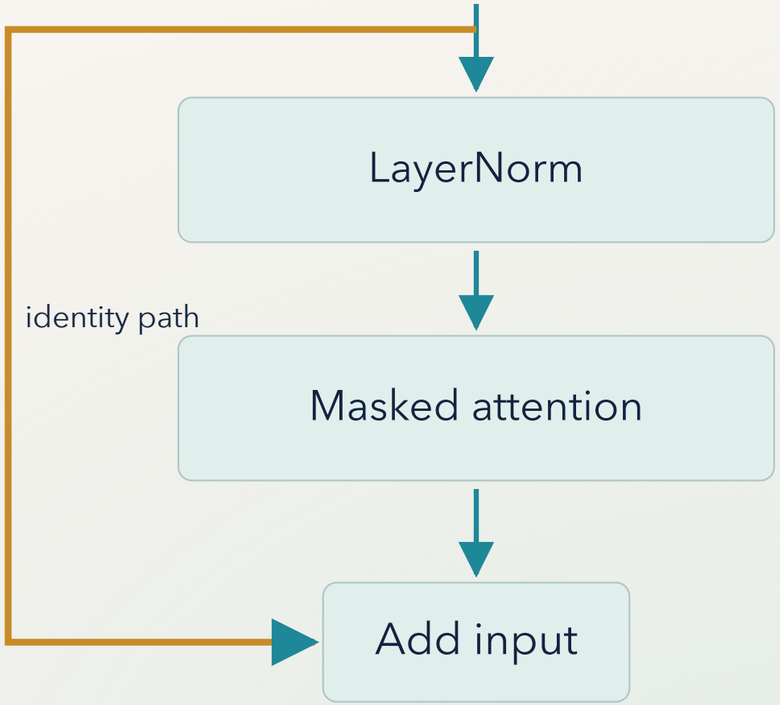


Mechanism diagram / illustrative teaching values

Normalization placement changes the block

Our teaching decoder normalizes before each sublayer.

> `x ← x + Attention(LN(x))`



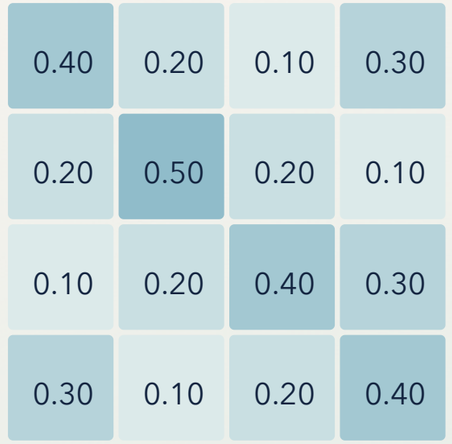
Then: normalized feed-forward update + residual

Mechanism diagram / illustrative teaching values

An encoder uses bidirectional context

Encoder self-attention can access all valid input positions.

Rows: queries / receiving positions

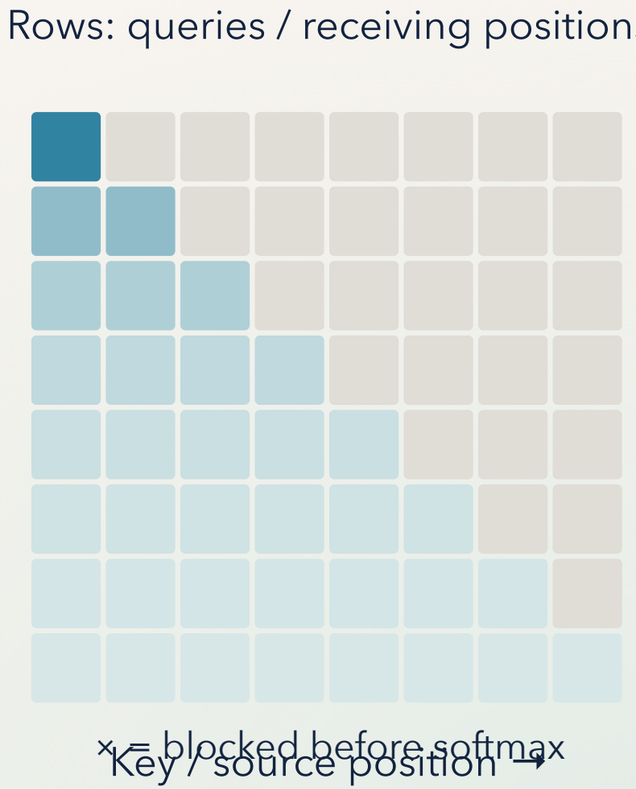


Key / source position →

Mechanism diagram / illustrative teaching values

A decoder-only model hides the future

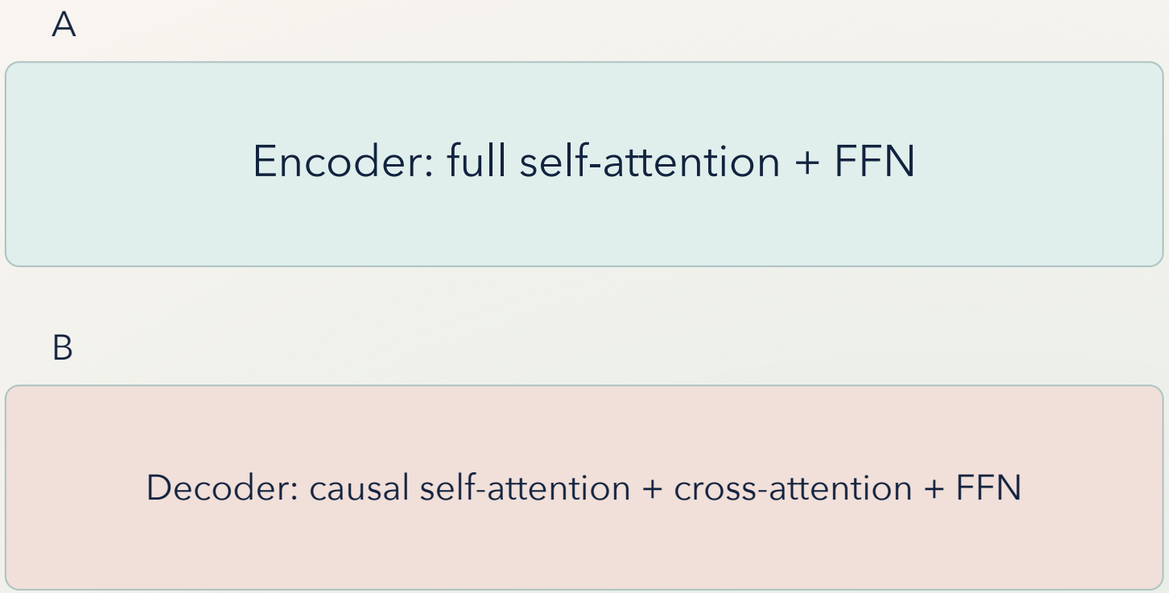
Masked self-attention supports next-token prediction from a prefix.



Mechanism diagram / illustrative teaching values

The original model had encoder and decoder

Decoder queries can retrieve from the separate encoder outputs.

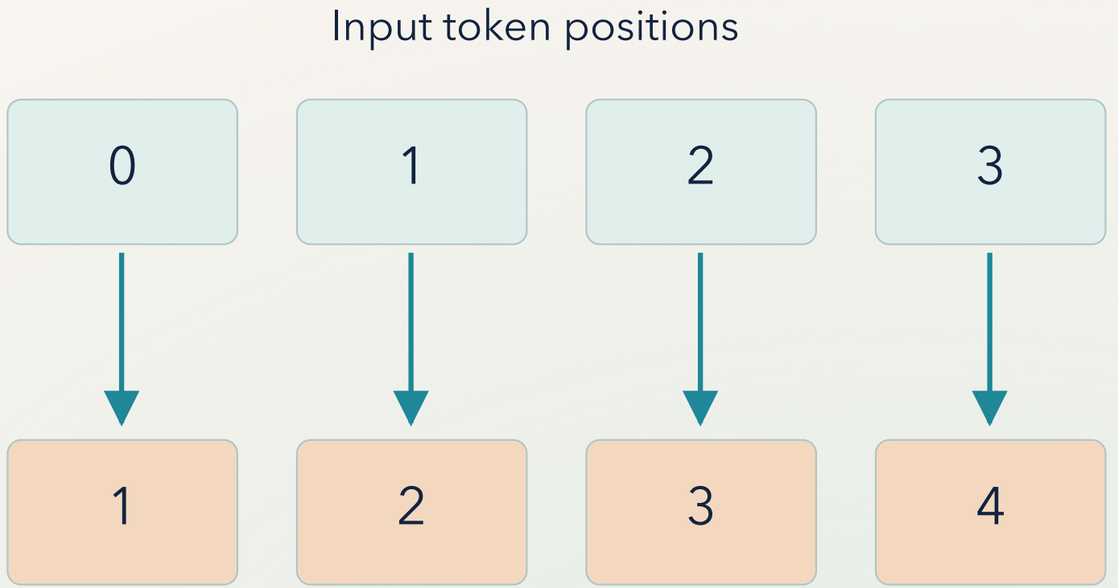


Mechanism diagram / illustrative teaching values

Predict the next token at every position

Shift targets one step ahead of inputs.

```
input: [0,1,2,3] target: [1,2,3,4]
```

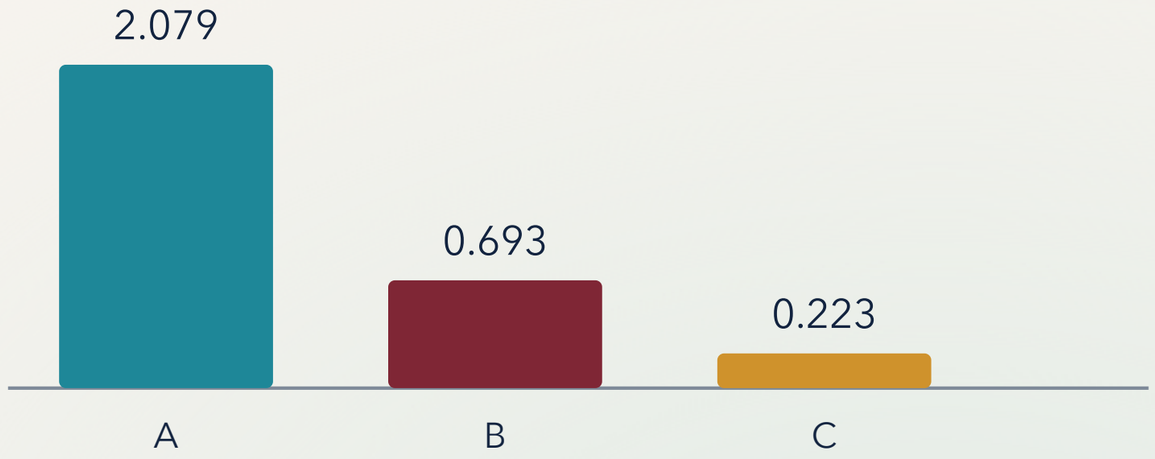


Mechanism diagram / illustrative teaching values

Cross-entropy scores the true next token

The model should assign high probability to observed next-token targets.

$$L = -\text{mean } \log p(y_t \mid x_{\le t})$$

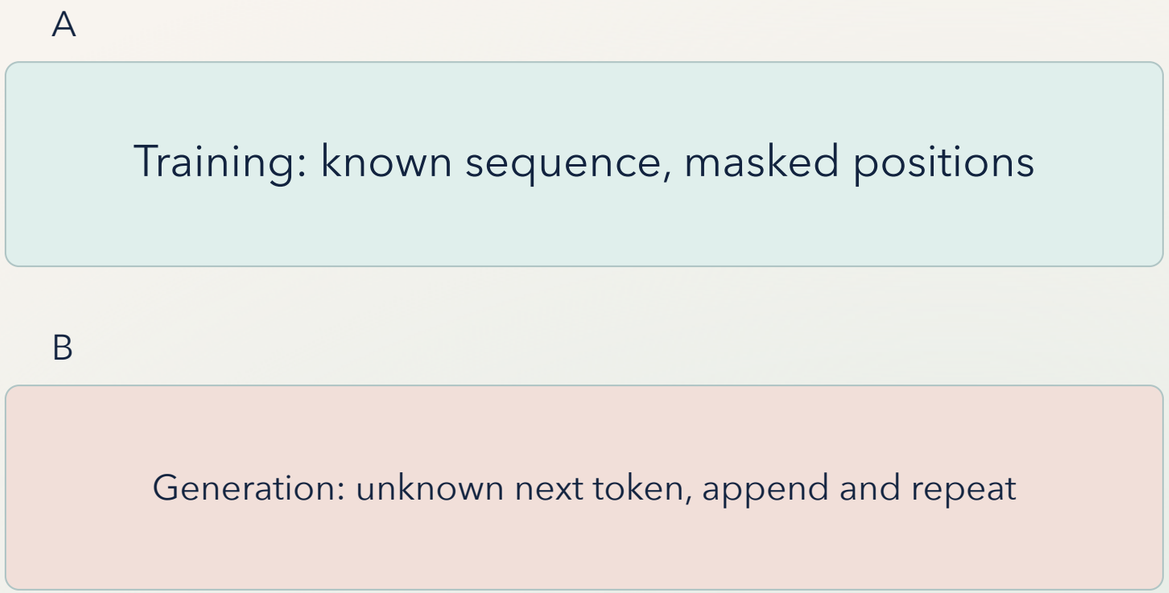


Illustrative values; see the stated calculation

Mechanism diagram / illustrative teaching values

Training is parallel across known positions

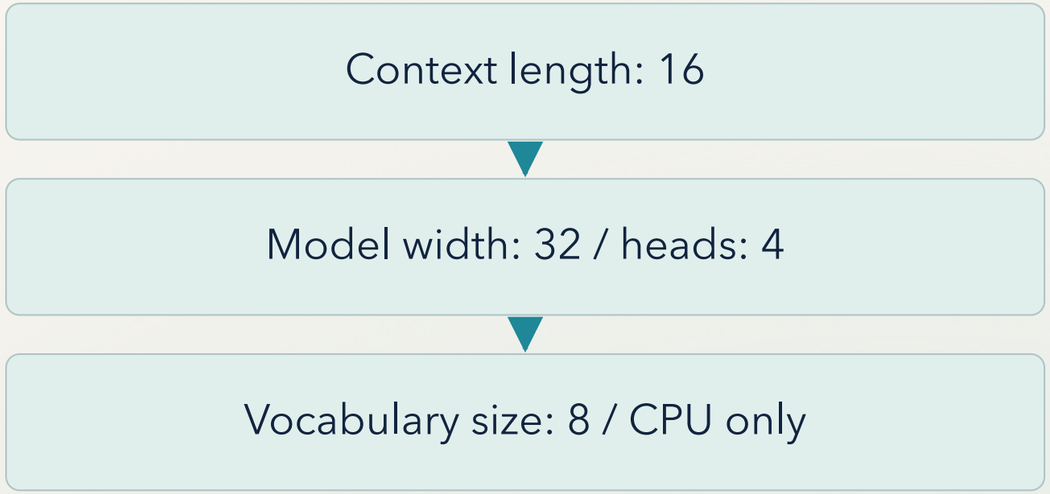
The causal mask permits computing many training losses together.



Mechanism diagram / illustrative teaching values

Open a tiny model you can inspect

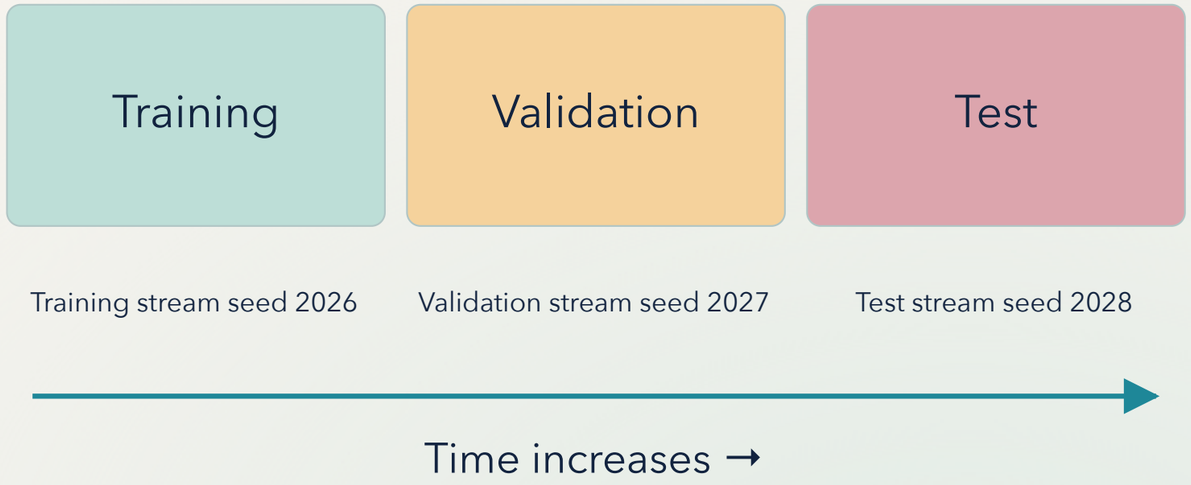
One block, width 32, four heads, and eight symbols keep the experiment small.



Mechanism diagram / illustrative teaching values

Define what generalization means here

Separate streams use different noise seeds but share the same synthetic process.



Mechanism diagram / illustrative teaching values

Read the four-dimensional score tensor

Every head produces a query-by-key grid for each example.

```
scores: (B, H, T, T)
```



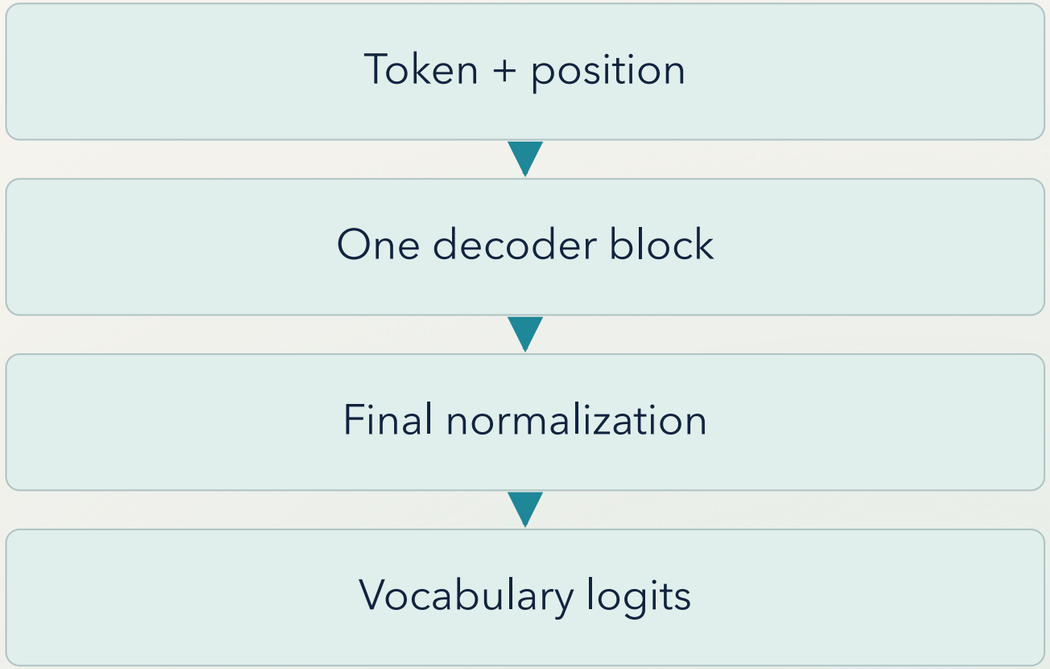
B = examples
H = attention heads
T × T = query/key pairs

Mechanism diagram / illustrative teaching values

Trace the model before training

The output has one vocabulary-logit vector per input position.

> `IDs: (B,T) → logits: (B,T,V)`



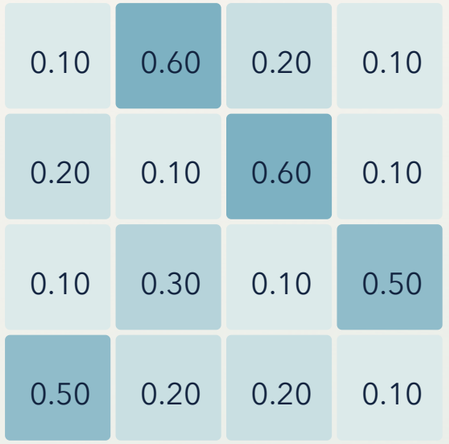
Mechanism diagram / illustrative teaching values

Fit a training-only bigram baseline

Estimate the next state from the current state using transition counts.

```
P(next | current) = smoothed count / row total
```

Rows: queries / receiving positions

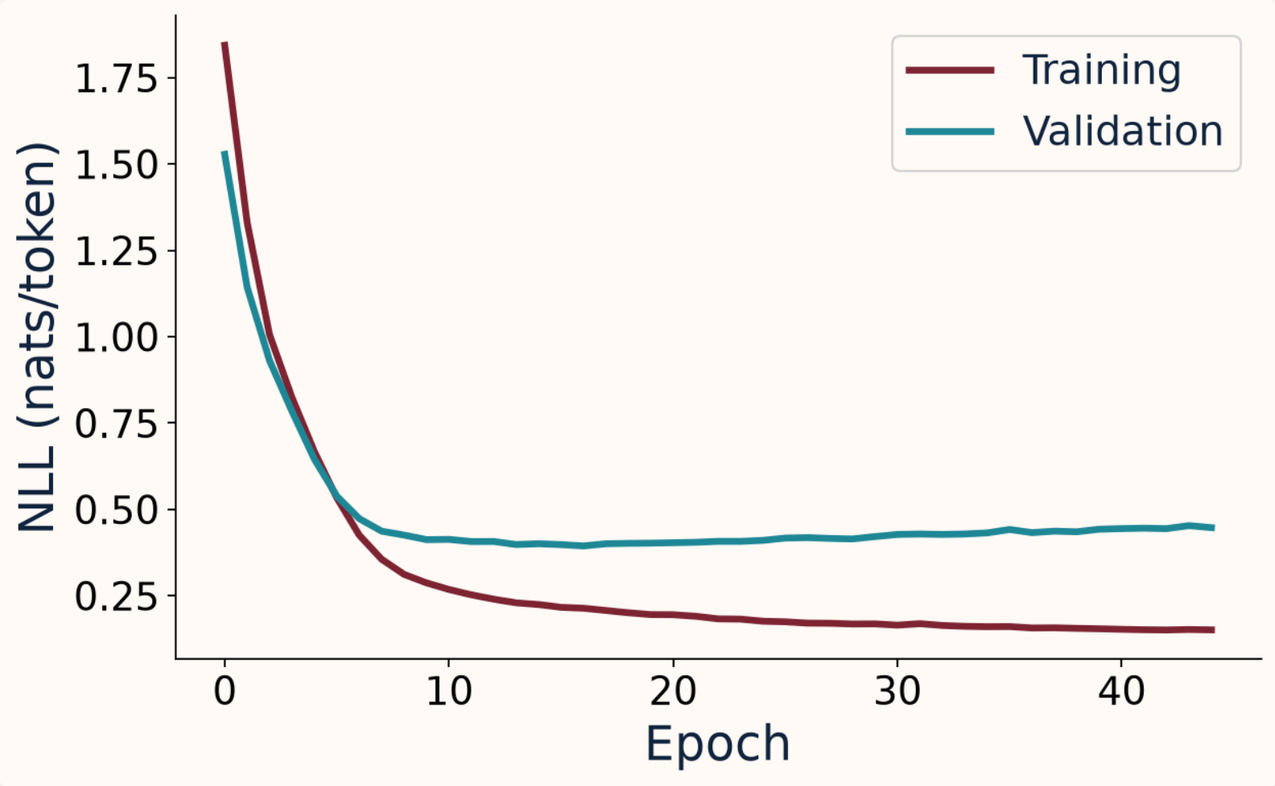


Key / source position →

Mechanism diagram / illustrative teaching values

Select by validation, evaluate test afterward

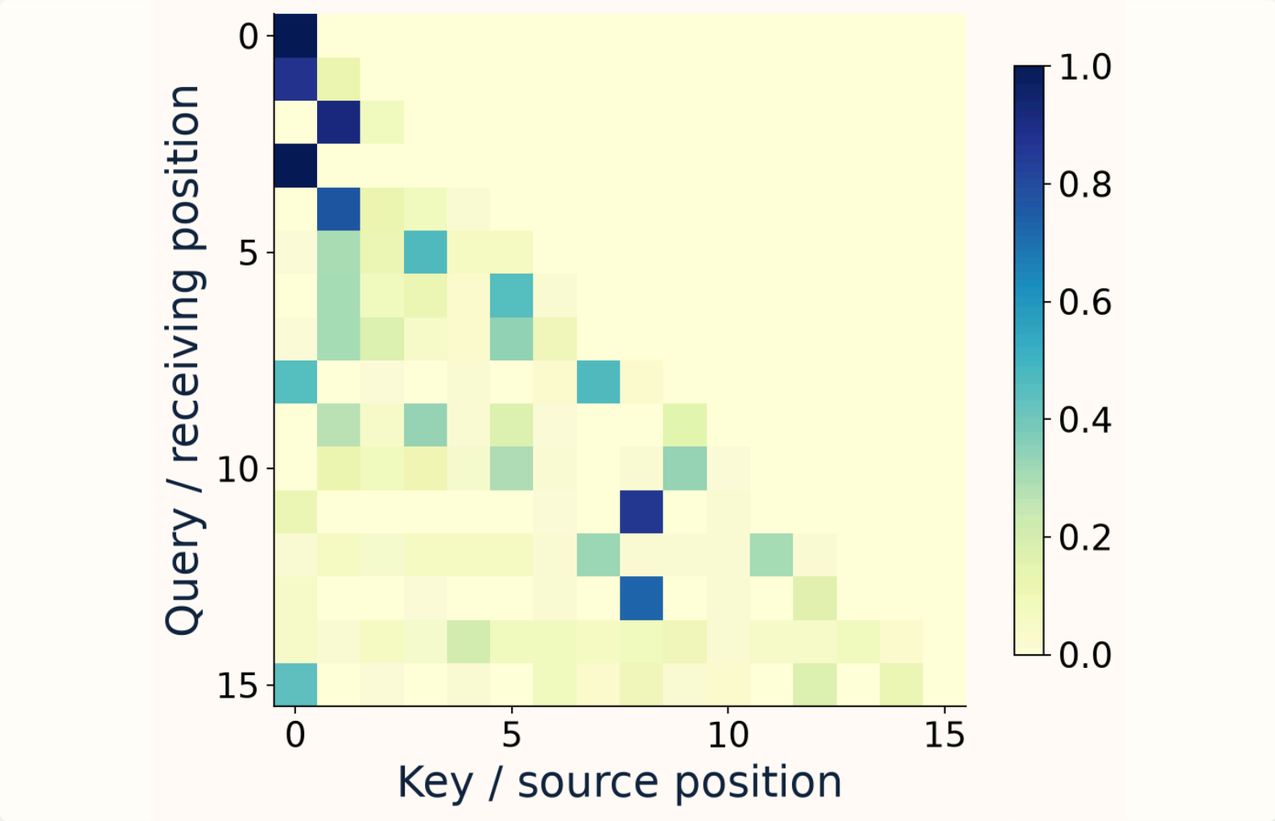
Keep the checkpoint with lowest validation negative log-likelihood.



Actual CPU notebook run · synthetic teaching data

Inspect actual learned attention

Rows are receiving positions; columns are source positions.



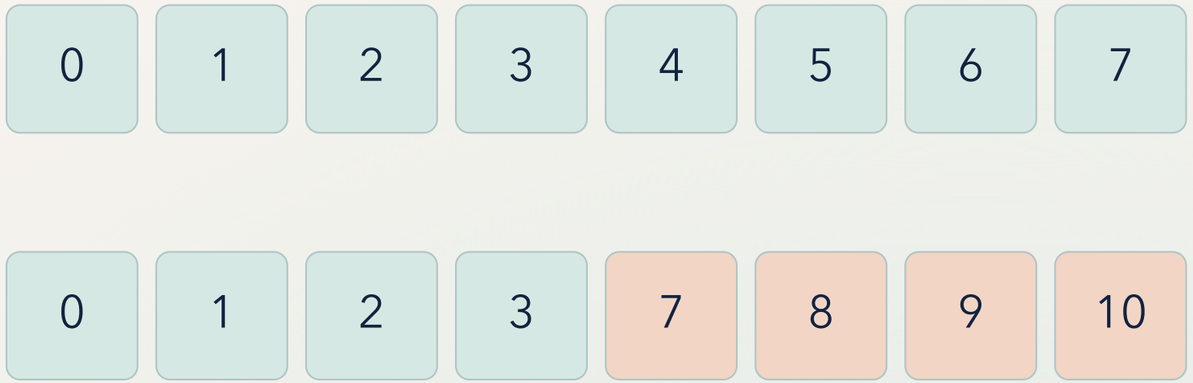
Actual CPU notebook run · synthetic teaching data

A future intervention must preserve earlier logits

Change tokens after position 7; logits at positions 0–7 should remain identical.

> `Δearlier = max |old_logits − new_logits|`

Original prefix versus modified future



Compare earlier logits: expected Δ = 0

Mechanism diagram / illustrative teaching values

Test the position claim precisely

Permute inputs to unmasked attention without positional information.

A

Permute input rows

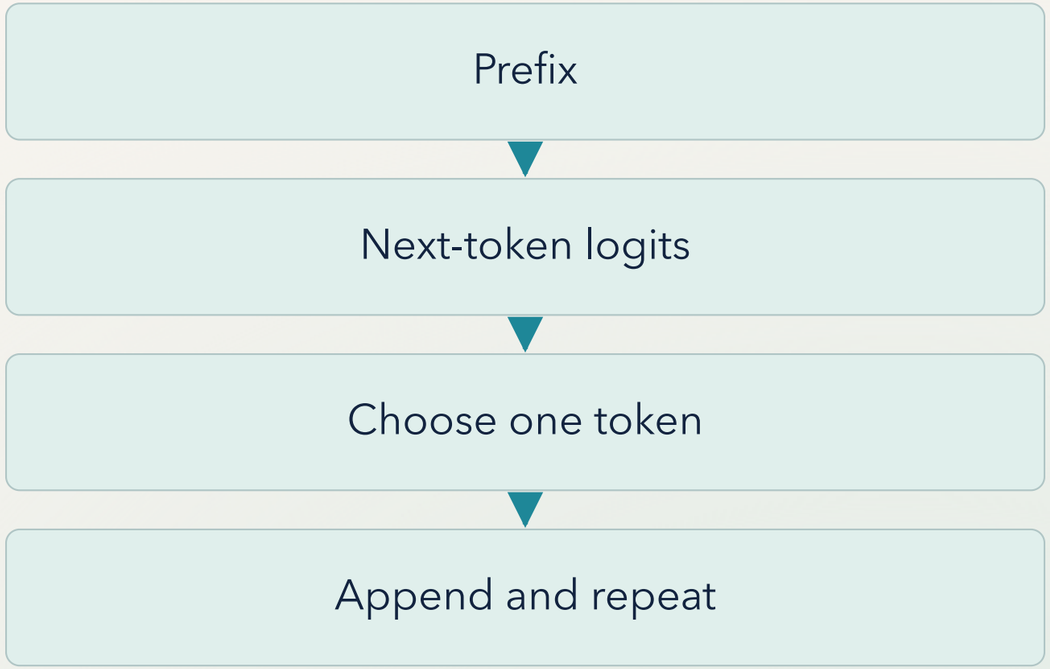
B

Output rows permute correspondingly

Mechanism diagram / illustrative teaching values

Generate one token, then append it

Autoregressive decoding grows the prefix sequentially.

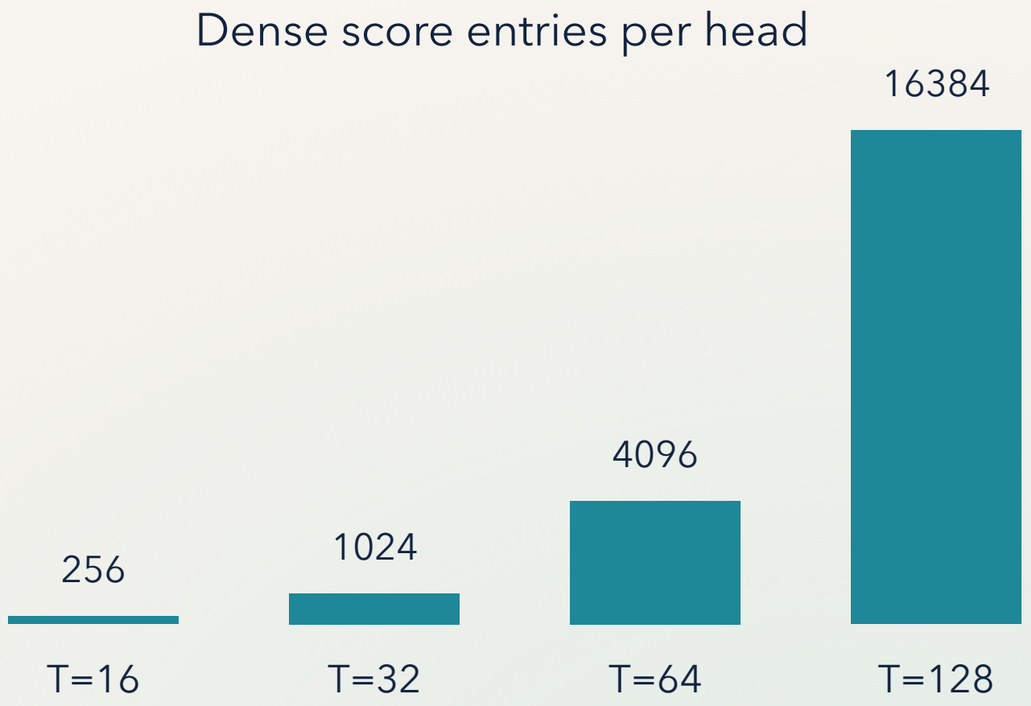


Mechanism diagram / illustrative teaching values

Long context has a measurable cost

Dense attention pair counts grow as T^2 at fixed head count.

> T=16: 256 pairs T=32: 1024 pairs



Mechanism diagram / illustrative teaching values

Choose models using task evidence

Streaming state, context length, data size, and compute budget all matter.

A

LSTM: recurrent state and sequential updates

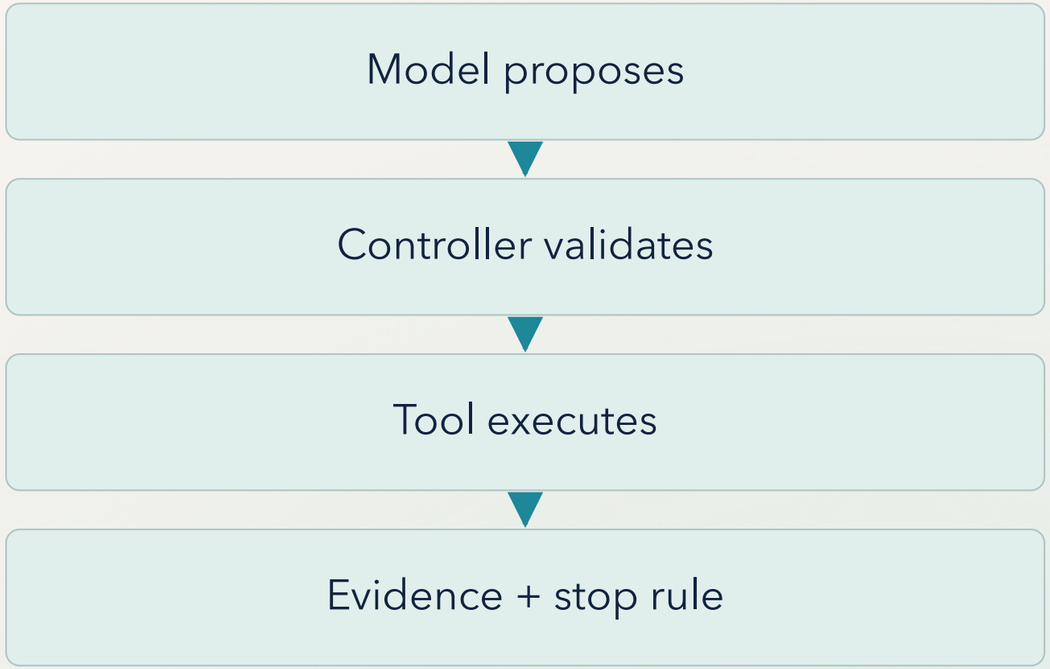
B

Transformer: attention context and parallel training

Mechanism diagram / illustrative teaching values

A token generator is one agent component

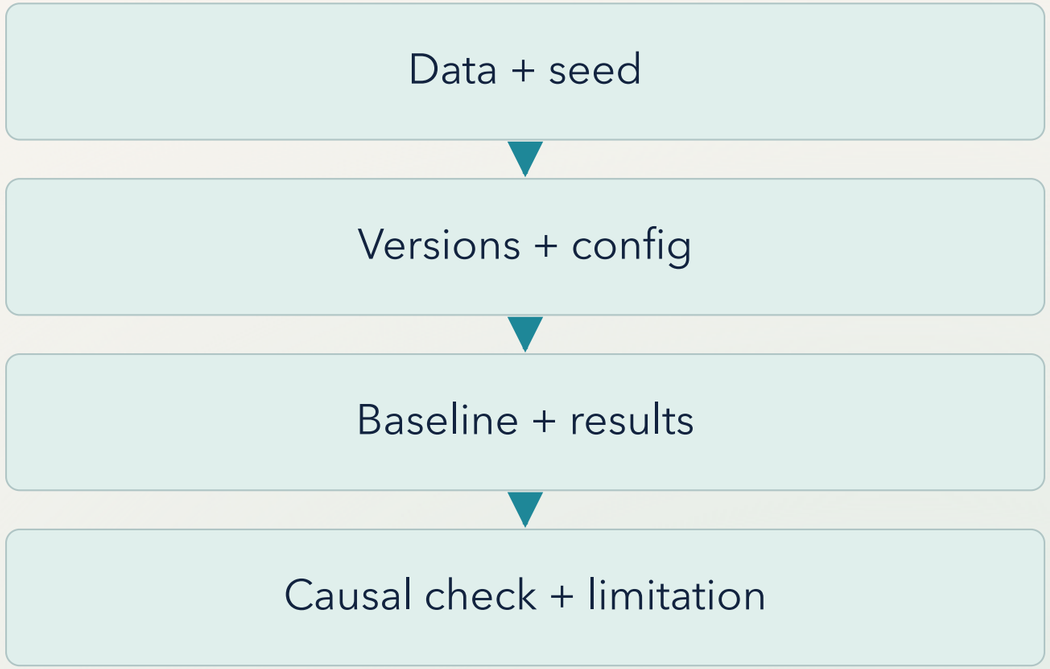
An agent also needs tools, state management, permissions, and a stopping rule.



Mechanism diagram / illustrative teaching values

Preserve the experiment, not just a score

Record data identity, configuration, baseline, checkpoint, and observed tests.

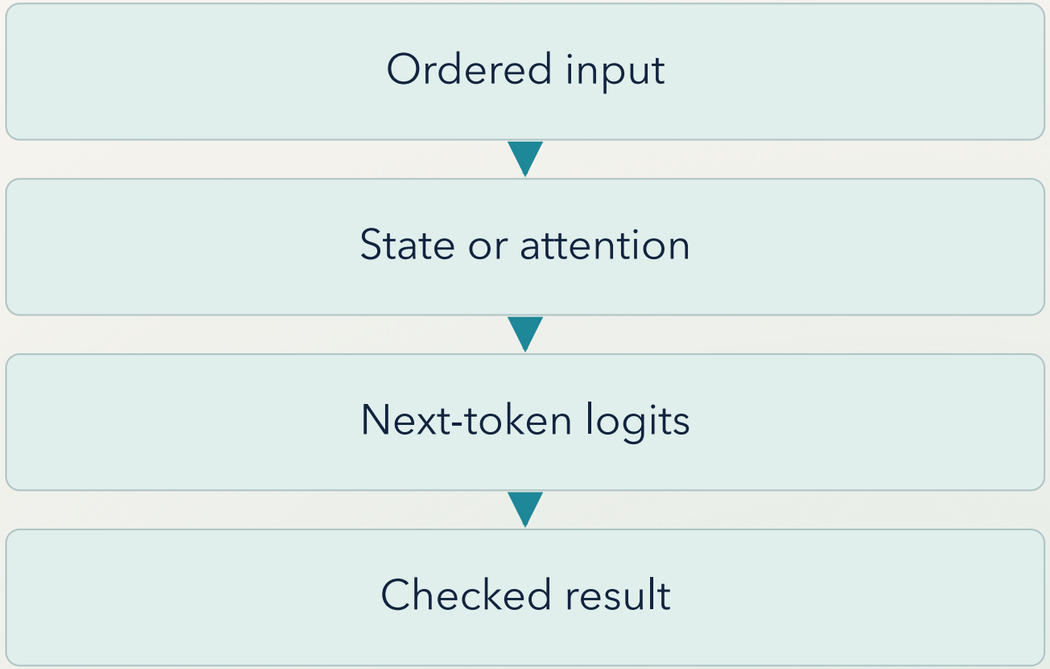


Mechanism diagram / illustrative teaching values

Exit: explain the complete causal chain

Sequence → memory → retrieval → representation → prediction → evidence.

Why is attention only part of the story?



Mechanism diagram / illustrative teaching values